

Oregon State University
College of Health
H524 Introduction to Biostatistics (AI-Enhanced)
Fall 2025

Credit Hours: 04

Room: online

Schedule:

Lecture: Monday, Wednesday 12:00-1:20pm

Lab: Wednesday: 8:00 am – 9:50 am

Course Instructor: Dr. John Molitor

Office Location: 153 Milam

Office Phone: 541-737-3834

E-Mail: john.molitor@oregonstate.edu

Office Hours: Wednesday, 10:00 am – 11:30 am

Course Description:

Quantitative analysis and interpretation of health data including probability distributions, estimation of associations, and significance tests such as chi-squared, one-way ANOVA, and linear regression.

AI-Enhanced Component: This course integrates generative AI tools as learning partners for statistical analysis. Students will learn to leverage AI for code generation, concept explanation, and problem-solving while developing critical skills in output verification and responsible AI use.

Academic Calendar

All students are subject to the registration and refund deadlines as stated in the Academic Calendar: <https://registrar.oregonstate.edu/osu-academic-calendar>

Instructor Communication

Please contact me privately Canvas inbox for matters of a personal nature.

Expectations for Time and Participation

This course combines approximately 120 total hours of instruction, online activities, and assignments for 4 credits.

Prerequisites

H 524 assumes basic knowledge of statistical concepts and procedures equivalent to that of an undergraduate statistics course, and MTH 105 (Introduction to Contemporary Mathematics) or MTH 111 (College Algebra) or a higher mathematics course.

Learning Resources

Resources include, but are not limited to:

- Lecture Notes
- Reading Assignments
- Discussions in class and on Canvas
- Problem Sets
- In-office help during office hours from both course instructor and lab instructors
- **AI Tools:** Access to ChatGPT, Claude, or GitHub Copilot (optional but recommended)

- **AI Prompt Templates:** Curated prompts for statistical analysis tasks

Textbook

Principles of Biostatistics, 2nd edition, 2022

Marcello Pagano and Kimberlee Gauvreau

Measurable Student Learning Outcomes

- Select and generate graphical and numerical summaries of data.
- Use principles of statistical inference to make conclusions about populations from samples.
- Communicate statistical findings to others.
- Use computer software to conduct statistical analysis.
- **AI Integration:** Effectively prompt AI tools for statistical tasks and critically evaluate their outputs.
- **Verification Skills:** Validate AI-generated code and results through independent checking.

Evaluation of Student Performance

Grading: Tests will be graded according to the following scale. It is possible, however, that adjustments will be made to this scale based on class performance.

Range	Grade
94–100%	A
89–93	A-
84–88	B+
79–83	B
74–78	B-
69–73	C+
64–68	C
59–63	C-
54–58	D+
49–53	D
44–48	D-
Less than 44	F

Evaluation: 30% homework (includes AI-enhanced problem sets), 30% midterm exam, 30% final exam, 10% class participation and AI reflection exercises.

Homework

Homework will be assigned regularly in class, approximately once a week. **AI Enhancement:** Each homework will include:

- Core statistical problems (70% of grade)
- AI verification task - validate AI-generated solutions (20% of grade)
- Brief AI usage documentation (10% of grade)

AI Use Policy

Permitted AI Use:

- AI tools are encouraged for learning, code generation, and concept exploration
- All AI-generated content must be verified and properly attributed
- Students must demonstrate understanding independent of AI assistance

Required Documentation:

- Cite AI tools used (name, version, date)
- Include key prompts for significant AI contributions
- Document verification steps taken

Academic Integrity:

- Submitting unverified AI output as your own work is academic dishonesty
- You are responsible for all errors in AI-generated content
- Exams will test your understanding with and without AI assistance

Course Schedule

The proposed schedule is subject to change by the instructor. What follows is a general outline of the proposed topics for discussion as well as the proposed dates. However, if we run out of time, get in a particularly lengthy discussion, or cover a topic more quickly than expected, there will likely be changes to the syllabus.

Week	Topic	Reading	AI Task
Week 0	Introduction to R Data Presentation Numerical Summary Measures	Pagano: Chapters 1, 2, 3	Configure AI tools; Generate basic EDA code
Week 1	Introduction to R (cont.) Data Presentation Numerical Summary Measures Rates and Standardization	Pagano: Chapters 1, 2, 3, 4	Compare AI plot suggestions; Document edits
Week 2	Probability	Pagano: Ch. 6 (6.1–6.4.2, 6.5)	Use AI for probability calculations; Verify with R
Week 3	Theoretical Probability Distributions Sampling Distribution of Mean	Pagano: Ch. 7 (7.1, 7.2, 7.4), 8	AI-scaffold CLT simulation
Week 4	Confidence Intervals	Pagano: Ch. 9	Generate CI code; Compare interpretations
Week 5	Hypothesis Testing Midterm exam	Pagano: Ch. 10	Critique AI p-value explanations
Week 6	Power Comparison of Two Means Analysis of Variance	Pagano: Ch. 11, 12	AI power analysis; Validate results
Week 7	Nonparametric Methods	Pagano: Ch. 13	When to use parametric vs non-parametric
Week 8	Catch-up from Weeks 6 & 7 Contingency Tables Multiple Contingency Tables	Pagano: Ch. 15, 16	AI for OR/RR interpretation
Week 9	Correlation Simple Linear Regression	Pagano: Ch. 17, 18	Generate diagnostic plots; Check assumptions
Week 10	Simple Linear Regression (continued)	Pagano: Ch. 18	Complete regression report with AI assistance

Final Exam: Available during finals week.

Lab Sessions - AI Integration

Wednesday labs will include:

- Traditional R programming exercises
- **AI-Enhanced Activities:**
 - Weeks 0-2: AI tool setup and basic prompting
 - Weeks 3-5: Using AI for statistical inference workflows
 - Weeks 6-7: AI-assisted method selection
 - Weeks 8-10: Building complete analyses with AI support

Exam Format Updates

- **Midterm (Week 5):**
 - Traditional closed-book section (50%)

- Open-resource section with AI permitted (50%) - must document AI use
- **Final Exam:**
 - Comprehensive
 - Mix of AI-assisted and independent work
 - Includes AI output evaluation questions

Diversity Statement

The College of Public Health and Human Sciences strives to create an affirming climate for all students including underrepresented and marginalized individuals and groups. Diversity encompasses differences in age, color, ethnicity, national origin, gender, physical or mental ability, religion, socioeconomic background, veteran status, sexual orientation, and marginalized groups. We believe diversity is the synergy, connection, acceptance, and mutual learning fostered by the interaction of different human characteristics.

Expectations for Student Conduct

The Student Conduct Code establishes community standards and procedures necessary to maintain and protect an environment conducive to learning, in keeping with the educational objectives of Oregon State University. This code is based on the assumption that all persons must treat one another with dignity and respect in order for scholarship to thrive. For the full Student Conduct Code see http://studentlife.oregonstate.edu/sites/studentlife.oregonstate.edu/files/code_of_student_conduct.pdf

Academic or Scholarly Dishonesty is prohibited and considered a serious violation of the Student Conduct Code. It is defined as an act of deception in which a Student seeks to claim credit for the work or effort of another person, or uses unauthorized materials or fabricated information in any academic work or research, either through the Student's own efforts or the efforts of another. For specifics related to offenses proscribed by the University see: <http://oregonstate.edu/studentconduct/offenses-0>

AI-Specific Academic Integrity: Using AI tools without proper attribution, submitting unverified AI output as original work, or using AI to fabricate data or references constitutes academic dishonesty.

Religious Holiday Statement

Oregon State University strives to respect all religious practices. If you have religious holidays that are in conflict with any of the requirements of this class, please see me immediately so that we can make alternative arrangements.

Students with Documented Disabilities

Accommodations for students with disabilities are determined and approved by Disability Access Services (DAS). If you, as a student, believe you are eligible for accommodations but have not obtained approval please contact DAS immediately at 541-737-4098 or at <http://ds.oregonstate.edu>. DAS notifies students and faculty members of approved academic accommodations and coordinates implementation of those accommodations. While not required, students and faculty members are encouraged to discuss details of the implementation of individual accommodations.